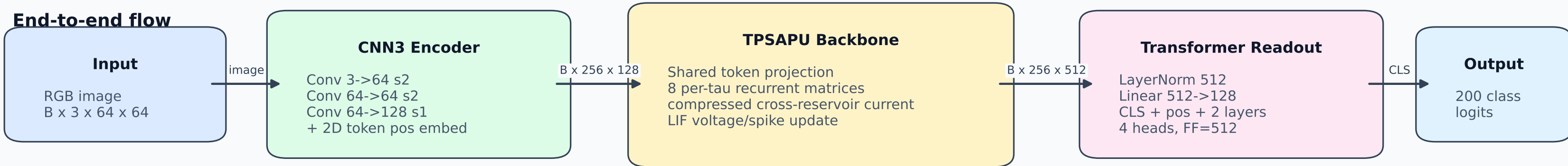


SAPU Tiny ImageNet Architecture: Per-Tau Recurrent + Compressed Cross-Reservoir Wiring

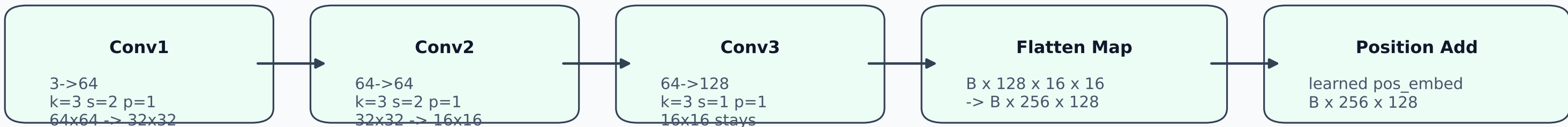
Command: `train_tiny_imagenet.py --backbone tpsapu_per_tau_cross_reservoir | encoder=cnn3 | decoder=membrane_transformer`

Tiny ImageNet defaults shown: image=3x64x64, tokens=16x16=256, embed_dim=128, taus=8, reservoir_dim=64, cross_rank=16, cross_gain=0.1

End-to-end flow

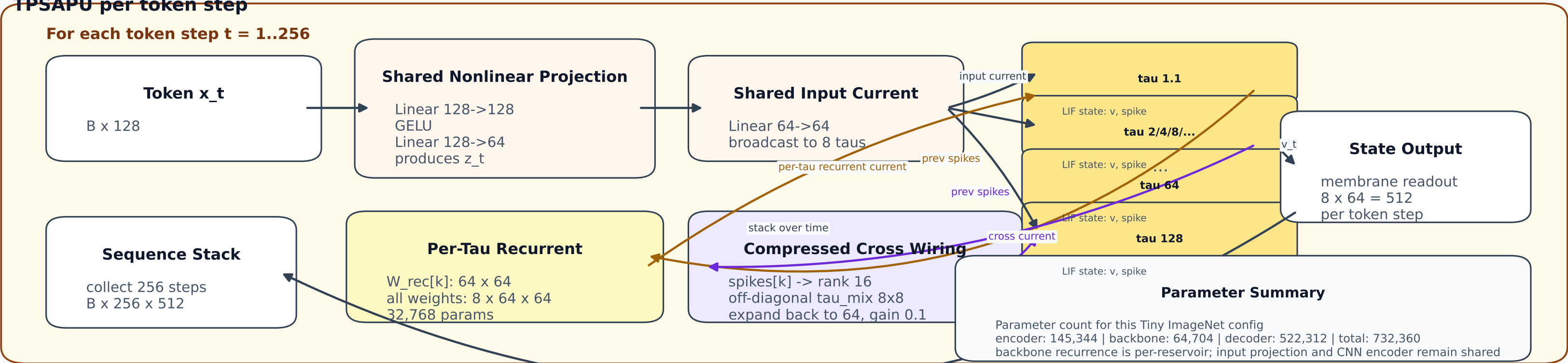


CNN3 tokenization

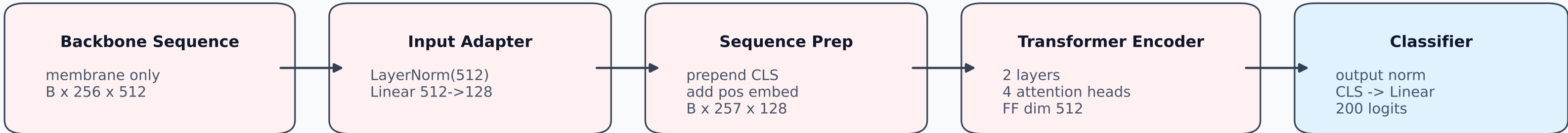


TPSAPU per token step

For each token step $t = 1..256$



Transformer readout



— forward dataflow — per-tau recurrent feedback — compressed cross-reservoir feedback